

Compound Interest

1. Simple and compound interest

Simple interest is paid only on the original principal.

Compound interest is paid on the principal plus interest already earned.

2. The compound interest formula

The balance after t years is $A = P(1 + r/n)^{nt}$, where P is the principal, r the annual rate and n the number of compounding periods per year.

3. Compounding frequency and time

More frequent compounding and longer time horizons both increase growth.

Doubling time is roughly 72 divided by the annual interest rate in percent.